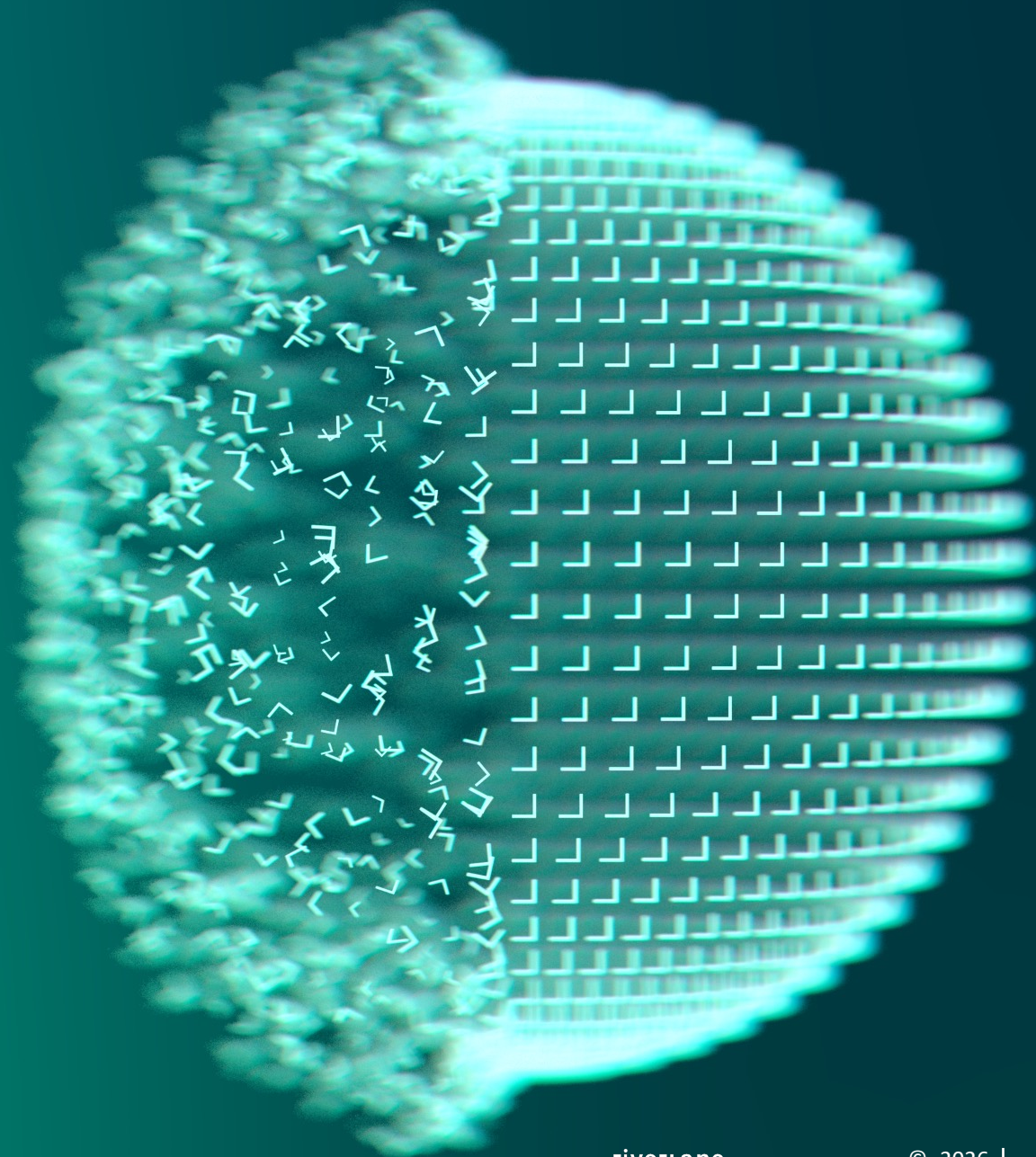




Electronic Structure Calculations in Quantum Chemistry

Jonny Davidson | Riverlane

QUICHE Tutorial, IEEE QCE26



Why Electronic Structure Matters

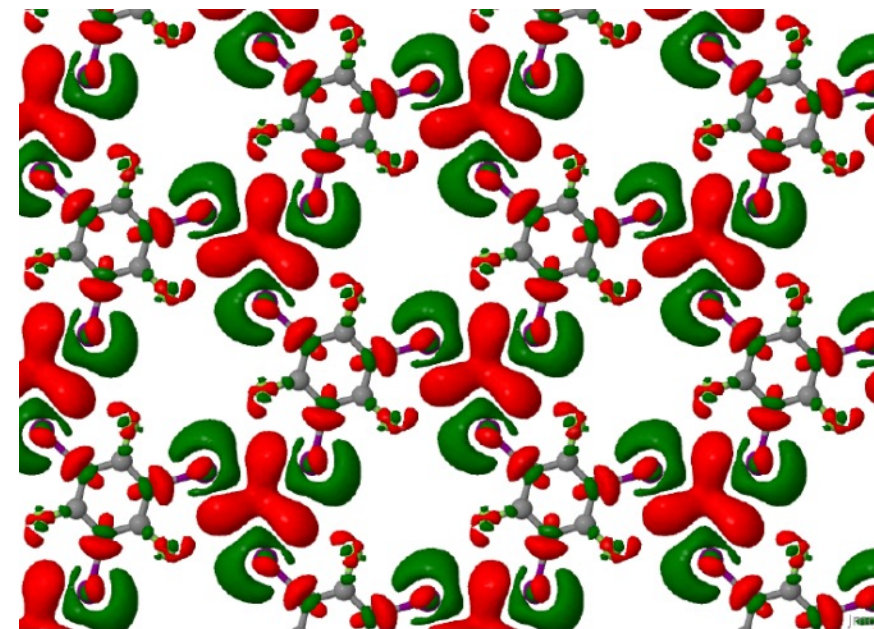
Every chemical property: bond length, reaction energy, spectra, catalytic activity... follows from the distribution of electrons around nuclei

Electronic structure theory is the toolkit for calculating that distribution and its energy

Industrial relevance:

- **Drug design:** predicting how molecules bind to protein targets
- **Catalyst discovery:** understanding reaction mechanisms at metal centres
- **Materials:** batteries, solar cells, novel semiconductors

What we want to calculate: where are the electrons, and how much energy does that configuration have?

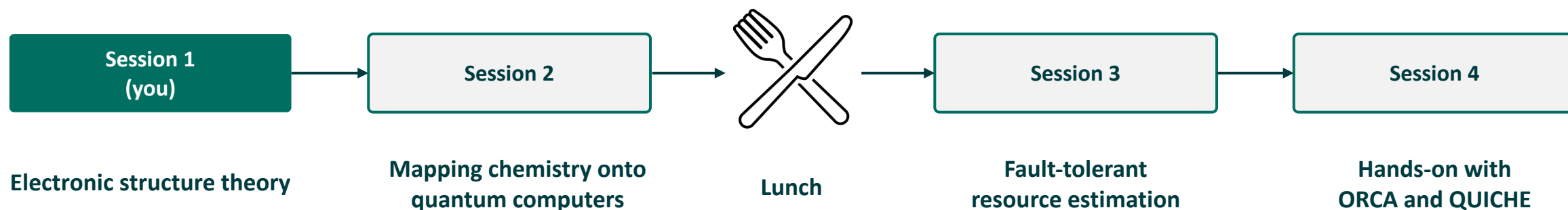


Electron density difference plot calculated for a monolayer of 1,3,5-Triiodo-2,4,6-trifluorobenzene. Red indicates accumulation of electron density, green depletion

Figure taken from Davidson J (2021) "Self-Assembly in Physisorbed Monolayers via Directional Intermolecular Interactions"

Session Roadmap

Before we can talk about electronic structure on quantum computers, we need to be understand the background of the problem we're trying to solve



The rationale:

- To understand why quantum computers might eventually help with chemistry, you first need an honest, precise picture of what classical methods can and cannot do.
- Every later session builds directly on the concepts introduced here.
- The aim is a working foundation for the sessions that follow, not a complete course in electronic structure theory

The QUICHE Project

QUICHE is a software pipeline linking the established computational chemistry software ORCA to quantum computing resource estimation from the quantum industry

The collaboration:

Quantum Motion: Full-stack silicon spin quantum computing company; contribute software and hardware expertise

FACCTs: Developers of ORCA, a major computational chemistry package (>100,000 users industry + academia) QUICHE integrates with as its chemistry front-end.

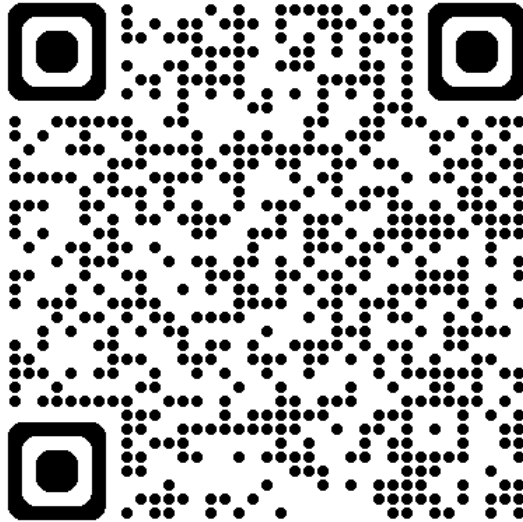
Riverlane: Quantum error correction specialists; contribute fault-tolerant algorithm expertise.

Funded by an 18-month UK–German bilateral quantum research grant.

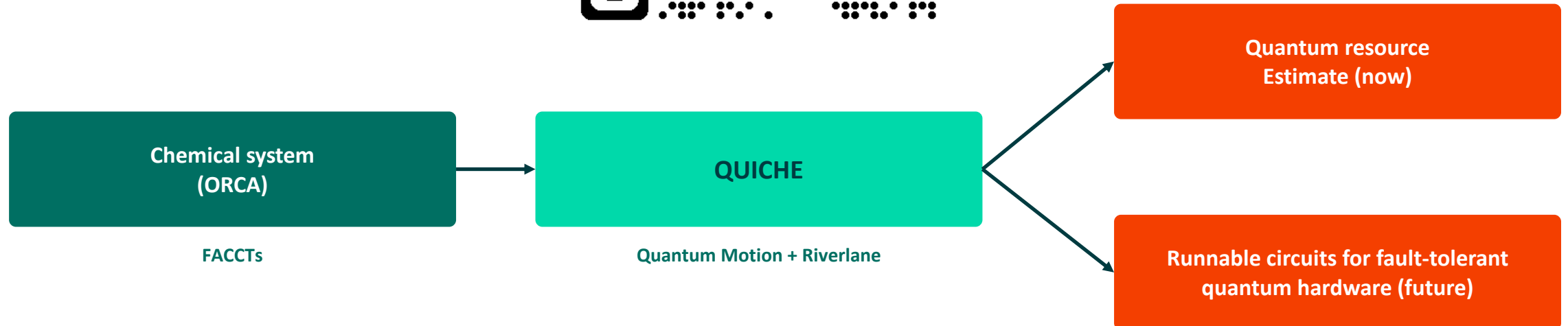


The QUICHE Project: link to release

Link to Github below (beta release)



Or:
`pip install pyquiche`



A starting point: the Born–Oppenheimer approximation

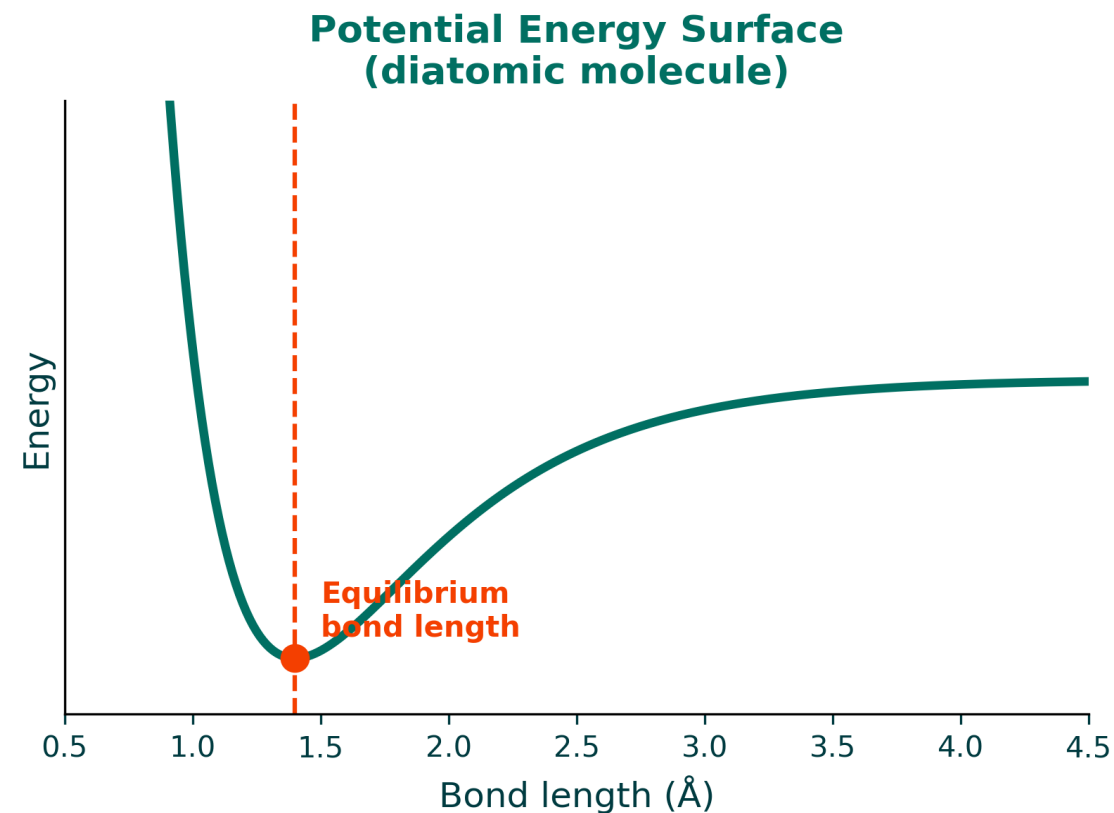
Proton-to-electron mass ratio is $\sim 1836:1 \rightarrow$ nuclei move far more slowly than electrons

Treat nuclei as fixed, classical point charges while solving for electrons

The resulting workflow:

- Solve the electronic problem at a fixed nuclear geometry
- Repeat at many geometries to map out the potential energy surface (PES)
- Use the PES to study vibrations, reaction paths, transition states etc.

Breaks down for near-degenerate electronic states (e.g. photochemistry) but otherwise a mainstay of computational chemistry



Problem Statement — The Electronic Schrödinger Equation

Goal: solve the time-independent Schrödinger equation at a fixed nuclear geometry (Born-Oppenheimer)

$$\hat{H}_{el} \Psi(r; R) = E(R) \Psi(r; R)$$

with the n-electron Hamiltonian:

$$\hat{H}_{el} = \overbrace{-\frac{1}{2} \sum_i \Delta_i}^{\text{Electron kinetic energy}} + \underbrace{\sum_{\{i,j>i\}} \frac{1}{|r_i - r_j|}}_{\text{Electron-nuclear attraction}} - \underbrace{\sum_{\{i,I\}} \frac{Z_I}{|R_I - r_i|}}_{\text{Nuclei-nuclei repulsion}} + \sum_{\{I,J>I\}} \frac{Z_I Z_J}{|R_I - R_J|}$$

↑ e-e repulsion

Convention:
i,j lower case letters refer to electrons
I,J upper case letters refer to Nuclei treated as fixed due to Born-Oppenheimer

Without the electron-electron term, this can be separated into n independent one-electron problems and is trivial. That term is the entire difficulty of quantum chemistry.

Solving the time-independent Schrödinger equation for a hydrogen atom

$$\hat{H}_{el} = -\frac{1}{2} \sum_i \Delta_i + \sum_{\{i,j>i\}} \frac{1}{|r_i - r_j|} - \sum_{\{i,I\}} \frac{Z_I}{|R_I - r_i|} + \sum_{\{I,J>I\}} \frac{Z_I Z_J}{|R_I - R_J|}$$

Only 1 electron

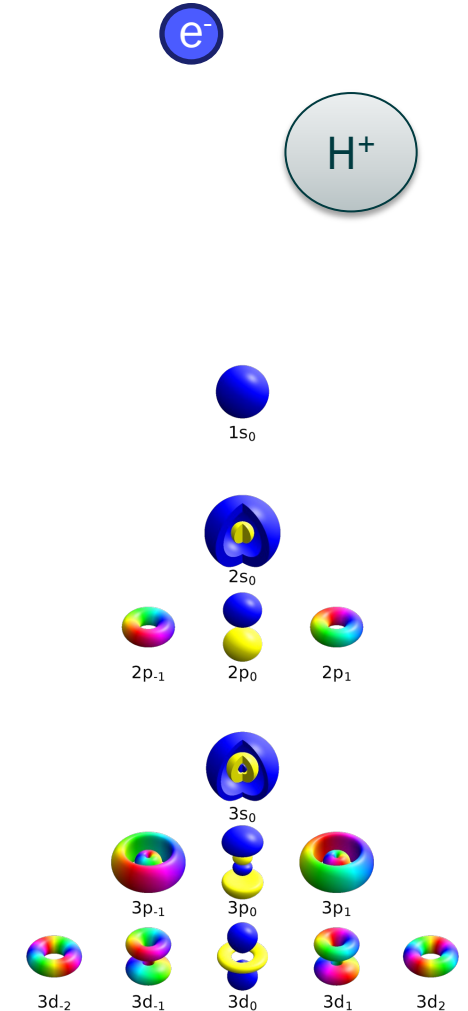
Only 1 nucleus

$$\hat{H}_{el} = -\frac{1}{2} \Delta - \frac{1}{r}$$



$$\Psi_{nlm}(r, \theta, \phi) = R_{n,l}(r) Y_l^m(\theta, \phi)$$

This leads to the familiar set of s, p and d orbitals defined by quantum numbers n, l and m

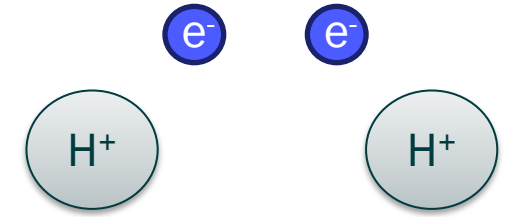


Convention:

All the above is in atomic units, setting terms such as charge and mass of the electron as well as fundamental constants to 1

Solving the time-independent Schrödinger equation for a hydrogen molecule

$$\hat{H}_{el} = -\frac{1}{2} \sum_i \Delta_i + \sum_{\{i,j>i\}} \frac{1}{|r_i - r_j|} - \sum_{\{i,I\}} \frac{Z_I}{|R_I - r_i|} + \sum_{\{I,J>I\}} \frac{Z_I Z_J}{|R_I - R_J|}$$



$$\hat{H}_{el} = -\frac{1}{2} \Delta_1 - \frac{1}{2} \Delta_2 + \frac{1}{|r_1 - r_2|} - \frac{1}{|R_1 - r_1|} - \frac{1}{|R_1 - r_2|} - \frac{1}{|R_2 - r_1|} - \frac{1}{|R_2 - r_2|} + \frac{1}{|R_1 - R_2|}$$

No exact analytical solution exists!

Time for some approximations!

The Self-Consistent Mean Field Approach: Hartree-Fock

The approximation:

- Instead of calculating **all** the pair-wise e-e interactions, assume each electron moves in the average field created by all the other electrons.
- An intractable many-body problem becomes a set of effective one-electron problems, solved iteratively
- The overall Ψ is assembled from these one-electron wavefunctions

Building Ψ from a set of one-electron orbitals (first attempt):

$$\Psi_0(x_1, \dots, x_n) \approx \phi_1(x_1)\phi_2(x_2)\cdots\phi_n(x_n) \quad (\text{Hartree product})$$

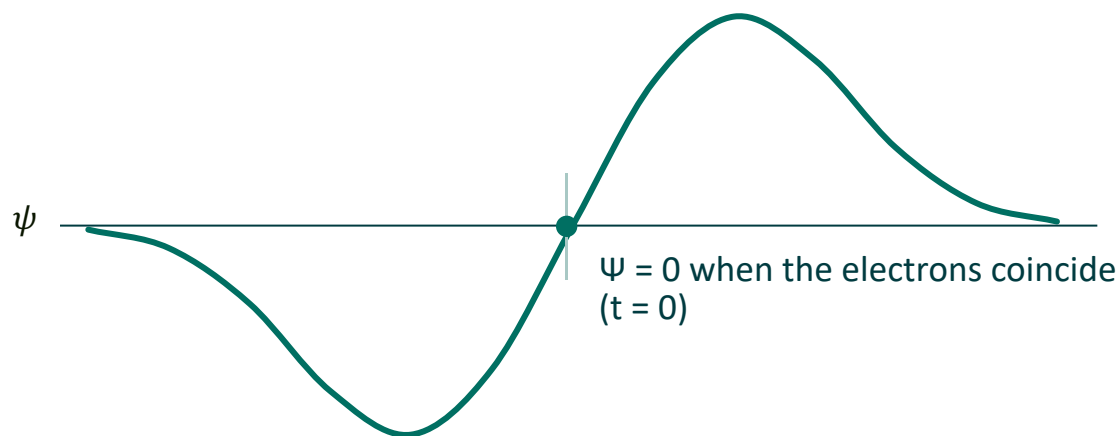
- Each spin orbital is single occupied and pairs a spatial orbital with a spin function: $\varphi_i(x) = \psi_i(r)\alpha$ (spin up) or $\psi_i(r)\beta$ (spin down).
- But a simple product is not antisymmetric (swapping two electron labels must flip the sign of Ψ), so this isn't quite the correct form...

Enforcing antisymmetry gives the Slater determinant (next slide);

Antisymmetry and the Fermi Hole

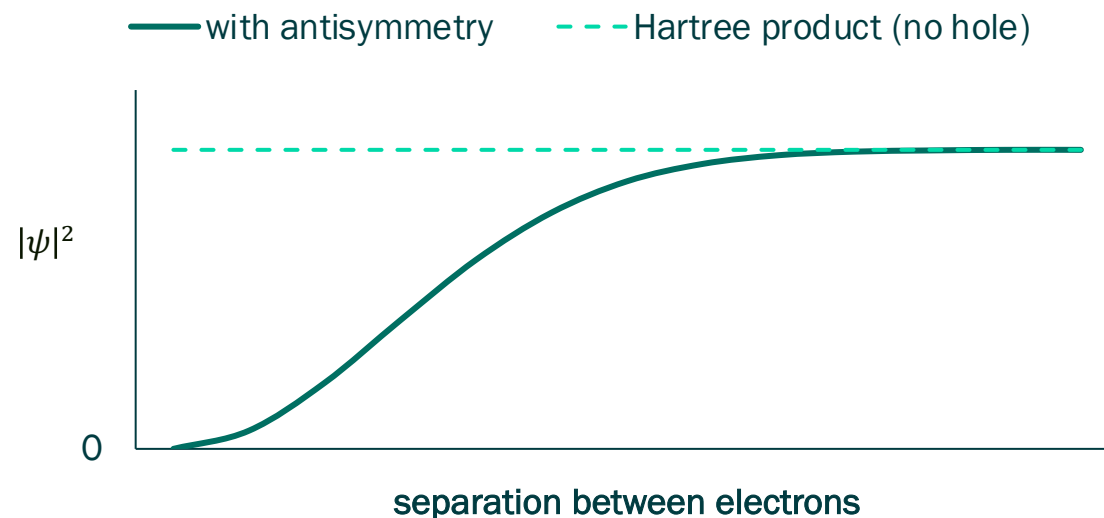
- For electrons with the same spin $\Rightarrow$ symmetric spin component $\Rightarrow$ the spatial wavefunction must be antisymmetric.
- Imagine a pair of electrons located around their midpoint c . Exchanging the pair is $c+t \rightarrow c-t$, so Ψ is an odd function of t : $\Psi(c-t) = -\Psi(c+t) \Rightarrow \Psi(t=0) = 0$.

Ψ along the relative coordinate (pair centred at c)



t = half the electron separation ($x_1 = c - t$, $x_2 = c + t$)

Same-spin pair density: the Fermi hole



- Antisymmetry forces same-spin electrons to avoid each other (the Fermi hole), giving less repulsion than a Hartree product predicts. That lowering is the exchange energy
- Opposite-spin pairs have no Fermi hole, so no exchange energy contribution

The Slater Determinant

The construction that satisfies antisymmetry — the Slater determinant

Two paired electrons sharing $\psi(r)$, with $\varphi_1(x) = \psi(r)\alpha$ and $\varphi_2(x) = \psi(r)\beta$:

$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}} \psi(r_1)\psi(r_2)[\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

Swap the two electron labels and the sign flips → Antisymmetric

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(x_1) & \chi_1(x_2) & \cdots & \chi_1(x_n) \\ \chi_2(x_1) & \chi_2(x_2) & \cdots & \chi_2(x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_n(x_1) & \chi_n(x_2) & \cdots & \chi_n(x_n) \end{vmatrix}$$

electron labels

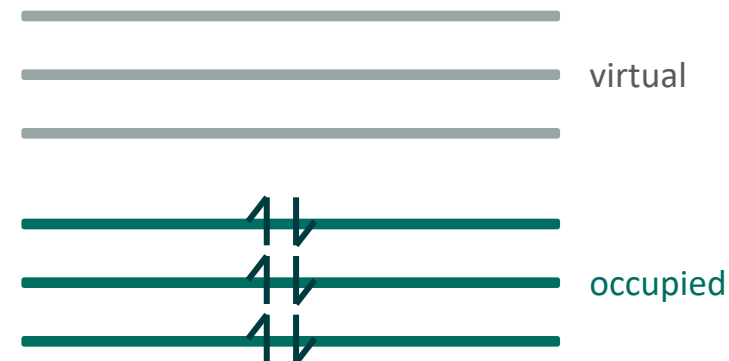
- **Rows = spin orbitals, columns = electrons**
- Swapping two electrons swaps two columns, so the determinant changes sign. Antisymmetry is built in!
- Two electrons in one spin orbital give identical rows, so it vanishes. Pauli exclusion for free

Basis Sets: Making the Equations Computable

We cannot solve for arbitrary functions, so instead each molecular orbital ψ is written as a weighted sum of m fixed, atom-centred basis functions χ :

$$\psi_i(r) = \sum_{\mu=1}^m c_{\mu i} \chi_{\mu}(r) \quad \Rightarrow \quad F c_i = \varepsilon_i S c_i$$

- We now solve for numbers — the coefficients c — rather than for functions.
- m basis functions give m spatial orbitals, i.e. $2m$ spin orbitals: the lowest N are occupied, the rest are virtual.
- Bigger basis, better answer, converging to the HF limit as $m \rightarrow \infty$



Basis sets in practice: Gaussians, in a few standard families:

Slater-type orbitals describe atoms better, but Gaussian-type orbitals are far easier to integrate...Gaussians are what everyone actually uses.

STO-3G — minimal: fast, qualitative only

6-31G*, 6-311++G** — split valence, with polarisation (*) and diffuse (+) functions added

cc-pVDZ, cc-pVTZ — systematically improvable, built for post-Hartree-Fock methods

def2-SVP, def2-TZVP — modern general-purpose workhorses

Hartree-Fock in Practice: the Self-Consistent Field Cycle

What we are solving for: the shape of each molecular spin orbital

The rule: choose the set of orbitals that gives the lowest total energy, keeping them orthonormal. Each one then obeys a single-electron equation:

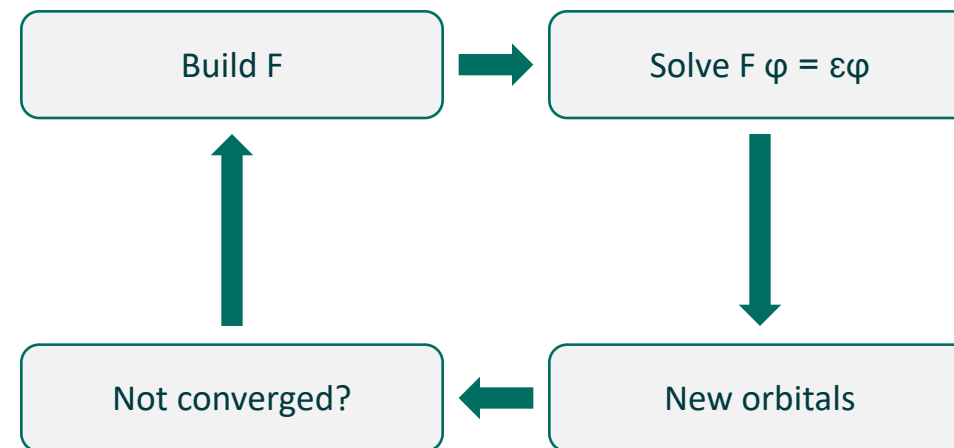
$$\hat{F}\phi_i = \varepsilon_i\phi_i$$

F is the Fock operator (one-electron Hamiltonian within the mean field) and ε_i is that orbital's energy.

F has three parts:

- **kinetic + nuclear:** the electron's motion and its attraction to the nuclei
- **Coulomb:** repulsion from the averaged charge cloud of all the other electrons
- **exchange:** a correction with no classical analogue, present only because Ψ must be antisymmetric

The catch: F is built from the mean field, which depends on the electron orbitals we are trying to find. Hence we repeat until the orbitals stop changing, the self-consistent field (SCF) cycle.



What Hartree-Fock Misses: Electron Correlation

HF is a single-determinant method: Coulomb repulsion is averaged into a mean field, while exchange is treated exactly.

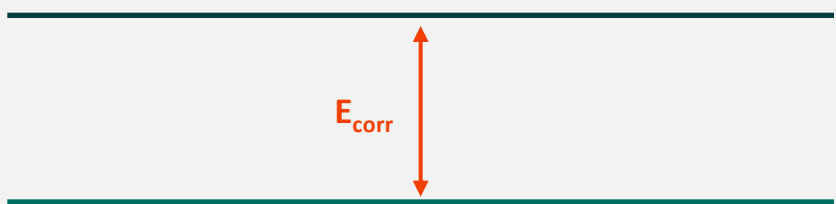
It is a variational upper bound, so any missing components are always stabilizing.

That missing piece is **Coulomb correlation**: electrons avoid one another more than a mean field predicts. It typically accounts for only 1–2% of the total energy.

But that 1–2% is the chemistry:

- Reaction energies and activation barriers are the same order of magnitude as the correlation energy itself
- Particularly relevant for strongly correlated systems (unpaired electrons, transition metals...)

Energy



$E(\text{HF})$ — mean-field upper bound

$E(\text{exact}) - E(\text{HF}) \sim 1\text{--}2\%$ of the total energy

$E(\text{exact})$ — includes correlation

Configuration Interaction

Add the additional determinants generated by exciting electrons into virtual orbitals:

$$\Psi_A = C_0^{(A)}\Psi_0 + \sum_{i,a} C_{ia}^{(A)}\Psi_i^a + \sum_{i,j,a,b} C_{ijab}^{(A)}\Psi_{ij}^{ab} + \dots$$

Same eigenvalue problem, now over the space of multiple determinants:

$$HC^{(A)} = EC^{(A)}$$

- **Why it works:** a larger variational space can only lower the energy, and the superposition lets electrons respond to each other's actual positions
- Considering all valid determinants (FCI) is exact within the basis; truncated CI is affordable but not size-consistent.



Full Configuration Interaction - Exact, and Out of Reach

No truncation at all. Include every determinant that can be built by distributing the n electrons over the m orbitals.

- Within a given basis, this is the **exact** solution of the electronic Schrödinger equation. It is the reference every other method is approximating.
- It is what defines the correlation energy: $E(\text{FCI}) - E(\text{HF})$, in the same basis.
- The determinant count is a binomial, not a polynomial:

$$N = C(m, n_{\alpha}) \times C(m, n_{\beta})$$

for m spatial orbitals and n_{α}, n_{β} electrons of each spin. Growing the active space makes the problem explode

Active space	Determinants
(4e, 4o)	36
(8e, 8o)	4,900
(14e, 14o)	11,778,624
(20e, 20o)	3.4×10^{10}
(50e, 50o)	1.6×10^{28}

This scaling speed is the reason we are here. A quantum computer stores the same space in one qubit per spin orbital 100 qubits for a (50e, 50o) active space, not 10^{28} amplitudes.

The Existing Benchmark: Coupled Cluster

Coupled cluster reaches a subset of the same excitations as FCI, but generates them from an exponential ansatz rather than a linear sum:

It does this using a cluster operator:

$$\Psi_{CC} = e^{\hat{T}} \Psi_0$$
$$\hat{T} = \sum_{i,a} t_i^a \hat{a}_a^\dagger \hat{a}_i + \frac{1}{4} \sum_{i,j,a,b} t_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i + \dots$$

First sum: singles. Second sum: doubles — hence CCSD.

Not variational — yet more accurate. CC gives no upper bound, but at the same truncation level it lands consistently closer to exact than CI, because the exponential form keeps the method size-consistent.

Why exponentiating helps

$$\hat{T}_2$$

doubles, included explicitly

$$\frac{1}{2} \hat{T}_2^2$$

quadruples appear for free, as products of doubles

CI truncated at the same level has to add those terms by hand — and doesn't.

CCSD(T) is the field's gold standard — but only for single-reference, weakly correlated systems. Where static correlation dominates (bond breaking, transition metals) it degrades badly

Density Functional Theory - A Parallel Framework

DFT is NOT a post-Hartree-Fock method — it is built on the electron density $\rho(r)$, not the wavefunction Ψ

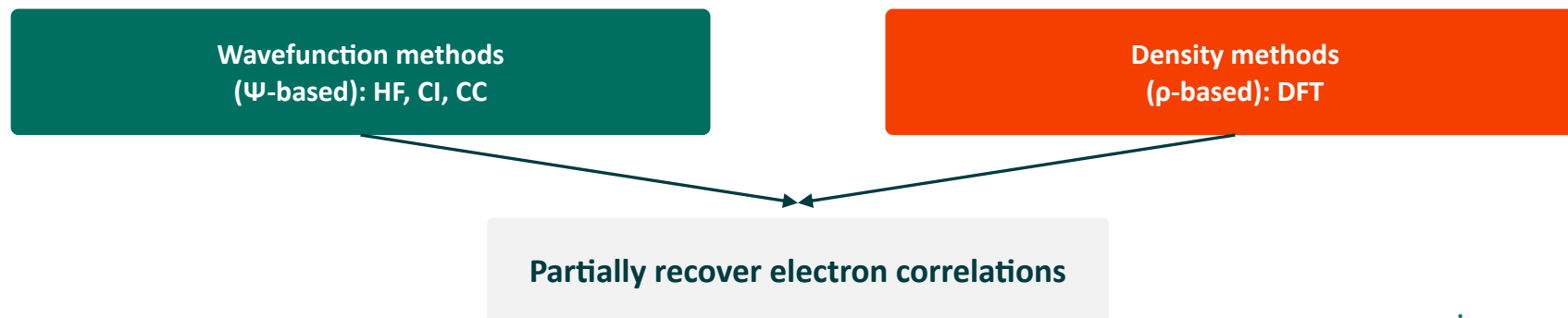
Hohenberg–Kohn: the ground-state energy is a unique functional of the density ρ alone, independent of Ψ — an existence proof only, with no recipe for the functional.

Neither $T[\rho]$ nor the non-classical part of $V_{\text{ee}}[\rho]$ is known in closed form

Kohn–Sham: a fictitious non-interacting system reproducing the real density. The unknowns collect into an exchange–correlation functional — formally exact; all approximation sits in the functional chosen:

$$\left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}(r) \right] \psi_i(r) = \varepsilon_i \psi_i(r)$$

- Cost comparable to HF; not systematically improvable; standard functions miss dispersion (corrections are available) but correlation remains a failure. The two branches aren't cleanly separated — Kohn–Sham reintroduces orbitals, and hybrid functionals borrow HF exchange outright.



The Method Hierarchy: Putting It All Together

Every wavefunction method trades accuracy against cost via how many determinants are included. DFT trades along a different axis — functional sophistication.

HF	One determinant — with the orbitals optimised	Mean-field. Exchange exact, no Coulomb correlation. $\sim N^4$.
CCSD	CCSD: uses cluster operator to add additional determinants	Most dynamic correlation, for single-reference systems. CCSD $\sim N^6$.
CCSD(T)	As CCSD, plus triples estimated perturbatively	Practical 'gold standard' — single-reference only. $\sim N^7$.
FCI	Every determinant the basis allows	Exact in the basis, but exponential — smallest systems only.
DFT (parallel branch)	No expansion — correlation folded into a functional	Cost $\sim N^3$ – N^4 , comparable to HF; not systematically improvable.

This whole ladder takes one determinant as a good starting point. When it isn't, such as for strongly correlated systems: bond breaking, near-degeneracy, transition metals... then only contributions from additional determinants can properly account for the complexity

Whichever method you use, often you only simulate a subset of active space

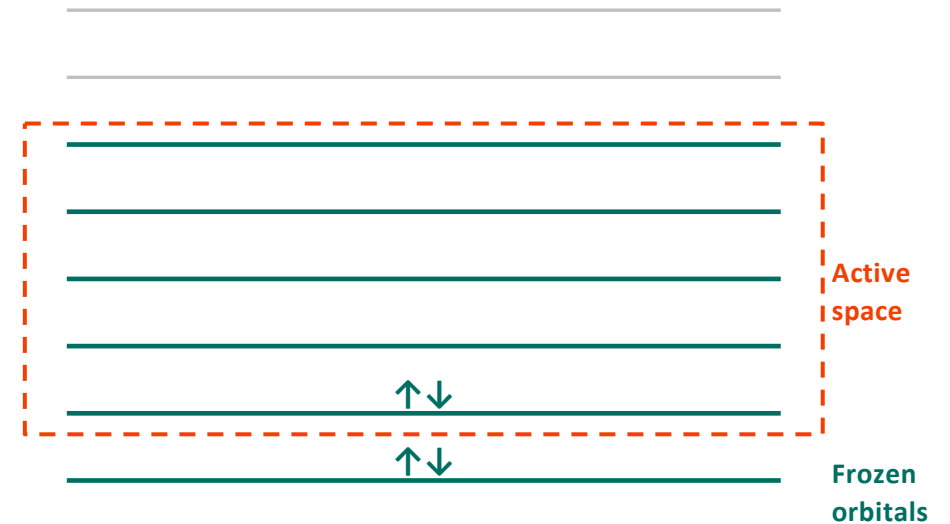
Treating all orbitals exactly is unaffordable → correlate only the ones that matter.

Choose an active space: n' , m' (n' electrons distributed among m' orbitals near the frontier.) Allow excitations only within that subset, and freeze the rest.

CASSCF: optimises the orbitals and the determinant coefficients together inside the active space. Jointly, this beats either optimisation alone.

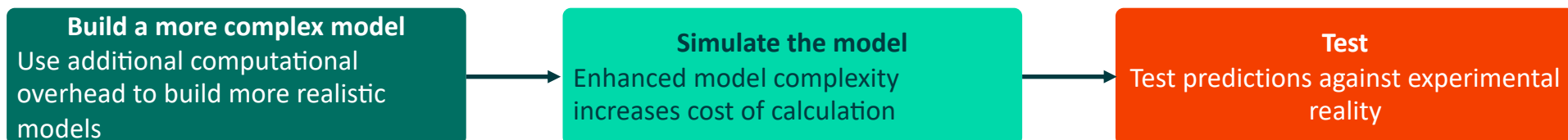
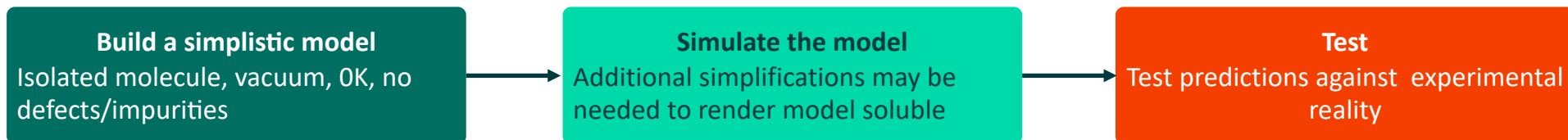
CASPT2: adds a perturbative correction for the dynamic correlation left outside the active space.

- **The catch:** choosing the active space is the most powerful and the most subjective step in a real calculation, guided by chemical intuition, orbital energies and natural orbital occupations.
- **Note: this is a wavefunction-side fix. A general multireference DFT remains an open problem.**



Why the size matters: the active space is where static correlation is treated exactly. Dynamic correlation from the frozen orbitals only comes back approximately — so the choice sets which correlation you get right, and how big the problem must be.

Implications for modelling: Why Chemists need Access to Quantum Now



The aim is not to simulate a spherical cow to greater accuracy, but to enable more accurate models to be built with fewer simplifications from the complicated real-world reality. A highly promising direction in quantum computing in chemistry is the integration of quantum solvers within larger-scale models (molecular dynamics, machine learning...)

Summary

What this session covered:

- The electronic Schrödinger equation, and why electron–electron repulsion is the main challenge
- Born–Oppenheimer separation of nuclear and electronic motion
- Hartree-Fock: the mean-field starting point, the Slater determinant, and exact exchange
- Basis sets and the LCAO approximation
- Post-HF wavefunction methods — configuration interaction and coupled cluster
- Density functional theory as an independent, density-based route to correlation
- Active spaces (CASSCF/CASPT2) as the practical route to spending cost only where the correlation matters

Classical methods trade accuracy against cost ($\sim N^4$ for HF to exponential for FCI) and for strongly-correlated systems that cost explodes with system size